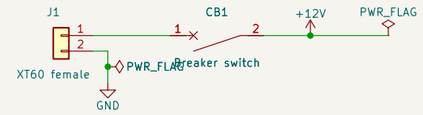
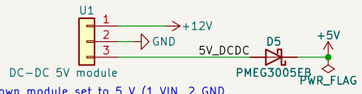
Power input and breaker switch



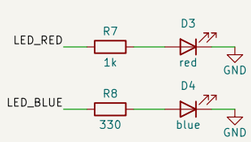
J1: female XT60 on the case, for the battery (LiFePO4 12.8 V 6 Ah with BMS) or a lab supply (12–14 V, current limit about 2 A).
The battery is charged off the mount.
CB1: resettable breaker switch, 3–5 A, rated for DC.

5 V supply for the ESP32 board



U1: DC–DC step–down module set to 5 V (1 VIN, 2 GND, 3 VOUT), off the board, wired to pads P5.
D5: USB and 12 V may be connected together; the ESP32 board then gets about 4.6 V from U1.

LEDs (on the case)



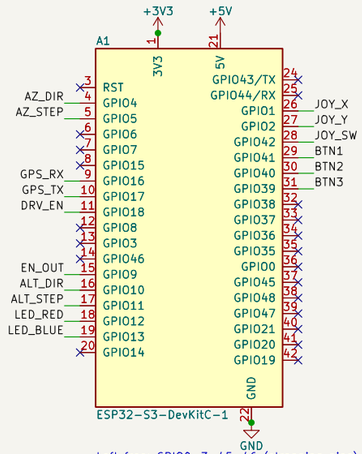
GPIO high = LED on.
PWM sets the brightness.

GPS (on the case)



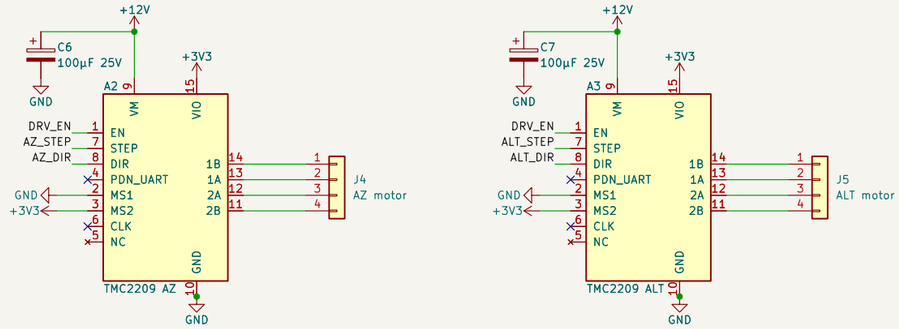
GPS module (u–blox NEO–6M): 1 VCC, 2 RX, 3 TX, 4 GND. VCC 3.3–5 V, 3.3 V logic, UART 9600 baud. Antenna towards the sky. Wired to pads P4.

Controller



Left free: GPIO0, 3, 45, 46 (strapping pins), GPIO19/20 (USB), GPIO35–37 (PSRAM on some boards), GPIO38 (RGB LED), GPIO43/44 (USB serial console).

Stepper drivers



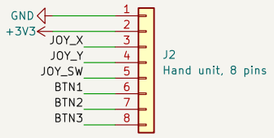
Standalone mode, no UART: MS1 = GND, MS2 = 3V3 gives 1/64 microsteps.
Motor current: potentiometer (Vref) on each module.
C6, C7 must sit close to the VM pins: they absorb the motor voltage spikes.

E–stop (on the case)



GPIO9 drives EN_OUT low to enable the drivers.
E–stop pressed or ESP32 in reset: R3 pulls DRV_EN high and both drivers switch off.
GPIO18 reads DRV_EN to detect it.

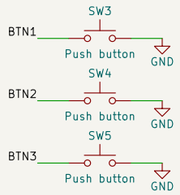
Hand–unit socket (case)



Hand unit (on the cable)



Analog joystick module (GND, VCC, VRx, VRy, SW), powered from 3.3 V.



Buttons and the joystick switch use the ESP32 internal pull–ups.

Notes

1. Supply, controller and drivers are all on the L: no cable crosses the azimuth joint.
2. +12V: 12.0–14.6 V from the battery or a lab supply. It feeds the drivers and U1.
3. TMC2209 header pin order differs between makers: wire by pin name. VIO at 3.3 V (ESP32 logic level).
4. Motor connectors: pins 1–2 one coil, 3–4 the other.
5. A1 pins: 1–22 = header J1, 23–44 = header J3.
6. Hand unit on an 8–wire cable (J2: for example GX16–8, or RJ45 with a network cable).
7. Keep the blue LED dim at night.
8. Board: stripboard (star–track.kicad_pcb). Parts on the case and the hand unit wire to pads P1–P5.

All electronics on the L of the mount
ESP32–S3 controller, two TMC2209 drivers, 12.8 V LiFePO4 battery or lab supply

Sheet: /
File: star–track.kicad_sch

Title: *Star–Track mount electronics*

Size: A3	Date: 2026–10–26	**Rev: 5**
KiCad E.D.A. 10.0.7		Id: 1/1